SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 2

SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING COURSE CODE: ECA09

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped- BL3,BL4,BL5**

Topics Covered:
Quantization- Truncation and Rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling.Parametric and Non parametric Spectral estimation. Decimation-Interpolation-Sampling rate conversion-Implementation of sampling rate conversion-Polyohase Decomposition -5G Decimation/Interpolation: For mmWave beamforming.

Assessment Tool number : CO3-AT2
Assessment Tool Weightage : 35%
Assessment Tool Name : Simulation Task
Duration of this assessment : 60 Minutes
Marks : 25 Marks
Type of Assessment Conduction : Self Learning (Home Task)

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question No	Conceptual Understanding 20	Model Design 25	Tool Usage 20	Analysis of Results 20	Documentation 10

QUESTION

6. Truncation vs Rounding Error Spectrum

Simulate the frequency-domain characteristics of quantization errors produced by truncation and rounding in a 20-tap FIR filter and compare their error spectra. (25) (BL5) (WK4)

Given Data:

● FIR Filter Length = 20 taps

● Input Signal = White Gaussian Noise

● Quantization = 8 bits

● FFT Length = 1024

Tasks:

1. Implement truncation-based quantization.
2. Implement rounding-based quantization.
3. Obtain error signals.
4. Compute FFT of error signals.
5. Plot error spectra.
6. Compare spectral characteristics.

INTRODUCTION

Quantization is an unavoidable process in digital signal processing where continuous-amplitude values are represented using a finite number of bits. The two most common quantization methods are truncation and rounding, each producing different quantization error characteristics. Truncation always removes the fractional part of a number, introducing a biased error, whereas rounding maps the value to the nearest quantization level, resulting in a more random and lower-power error. In FIR digital filters, these quantization errors affect the filter's output spectrum and overall performance. By analyzing the frequency-domain characteristics of the quantization errors using the Fast Fourier Transform (FFT), the spectral distribution of errors produced by truncation and rounding can be compared. This simulation helps in understanding how the choice of quantization method influences the noise spectrum and the quality of digital signal processing systems.

OBJECTIVE

- To implement a 20-tap FIR filter using White Gaussian Noise as the input signal.
- To perform 8-bit truncation-based and rounding-based quantization on the FIR filter output.
- To obtain and analyze the quantization error signals for both quantization methods.
- To compute the 1024-point FFT of the error signals and evaluate their frequency-domain characteristics.
- To compare the error spectra of truncation and rounding and identify the quantization method that provides better spectral performance.

TOOL USAGE

Software - MATLAB Online / Basic

MATLAB Functions Used

- **randn()** – Generates White Gaussian Noise
- **fir1()** – Designs a 20-tap FIR low-pass filter
- **filter()** – Filters the input signal
- **fft()** – Computes the Fast Fourier Transform
- **fftshift()** – Centers the FFT spectrum
- **abs()** – Computes magnitude
- **plot()** – Displays the spectra
- **subplot()** – Plots multiple graphs
- **grid on** – Adds grid lines
- **xlabel(), ylabel(), title(), legend()** – Graph labeling

CODE

```
%% Truncation vs Rounding Error Spectrum

clc;

clear;

close all;

%% Given Data

filterLength = 20;          % FIR filter length

N = 4096;                   % Number of samples

bits = 8;                   % Quantization bits

FFT_Length = 1024;          % FFT Length

%% Generate White Gaussian Noise

x = randn(1,N);

%% Design 20-Tap FIR Filter

h = fir1(filterLength-1,0.4);

%% Filter the Input Signal

y = filter(h,1,x);

%% Quantization Parameters

ymax = max(abs(y));

step = (2*ymax)/(2^bits);

%% -----------------------------

%% Truncation Quantization

%% -----------------------------

y_trunc = floor(y/step)*step;

%% -----------------------------

%% Rounding Quantization

%% -----------------------------

y_round = round(y/step)*step;
```

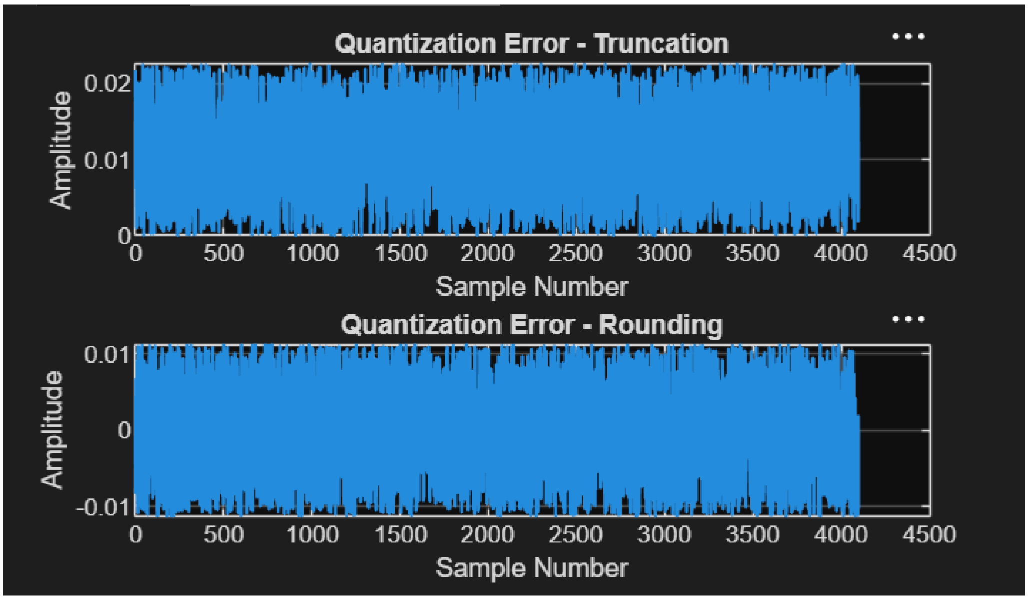
```
%% ----------------------------
%% Error Signals
%% ----------------------------
error_trunc = y - y_trunc;
error_round = y - y_round;
%% ----------------------------
%% FFT of Error Signals
%% ----------------------------
E_trunc = fftshift(fft(error_trunc,FFT_Length));
E_round = fftshift(fft(error_round,FFT_Length));
freq = linspace(-0.5,0.5,FFT_Length);
%% Magnitude Spectrum (dB)
Spec_trunc = 20*log10(abs(E_trunc)+eps);
Spec_round = 20*log10(abs(E_round)+eps);
%% ----------------------------
%% Plot Error Signals
%% ----------------------------
figure;
subplot(2,1,1)
plot(error_trunc,'LineWidth',1.2)
grid on
title('Quantization Error - Truncation')
xlabel('Sample Number')
ylabel('Amplitude')
subplot(2,1,2)
plot(error_round,'LineWidth',1.2)
grid on
```

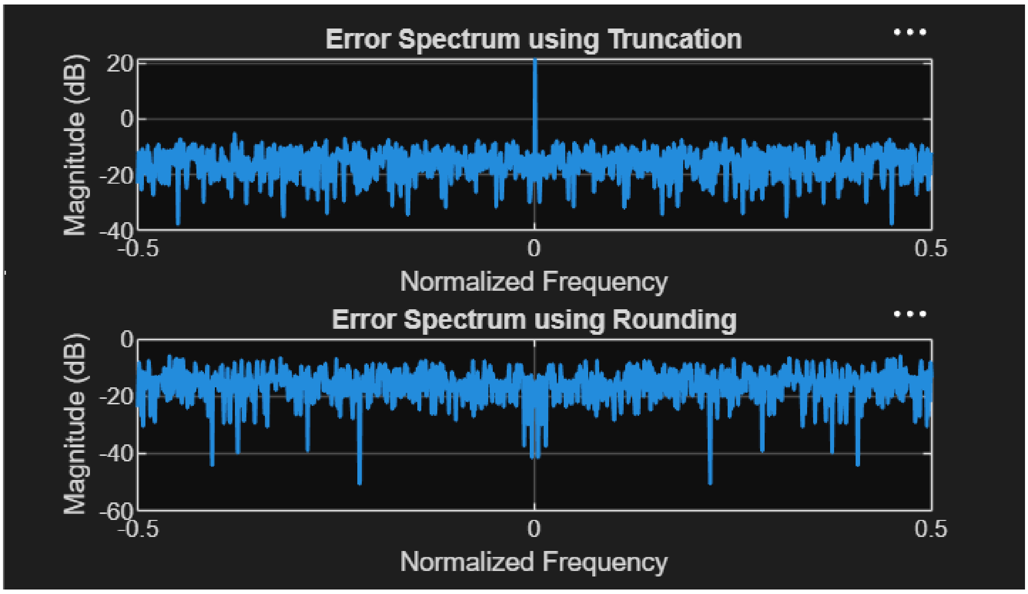
```
title('Quantization Error - Rounding')

xlabel('Sample Number')

ylabel('Amplitude')

%% -----------------------------

%% Plot Error Spectra

%% -----------------------------

figure;

subplot(2,1,1)

plot(freq,Spec_trunc,'LineWidth',1.5)

grid on

xlabel('Normalized Frequency')

ylabel('Magnitude (dB)')

title('Error Spectrum using Truncation')

subplot(2,1,2)

plot(freq,Spec_round,'LineWidth',1.5)

grid on

xlabel('Normalized Frequency')

ylabel('Magnitude (dB)')

title('Error Spectrum using Rounding')

%% -----------------------------

%% Comparison Plot

%% -----------------------------

figure;

plot(freq,Spec_trunc,'r','LineWidth',1.5);

hold on;

plot(freq,Spec_round,'b','LineWidth',1.5);

grid on;
```

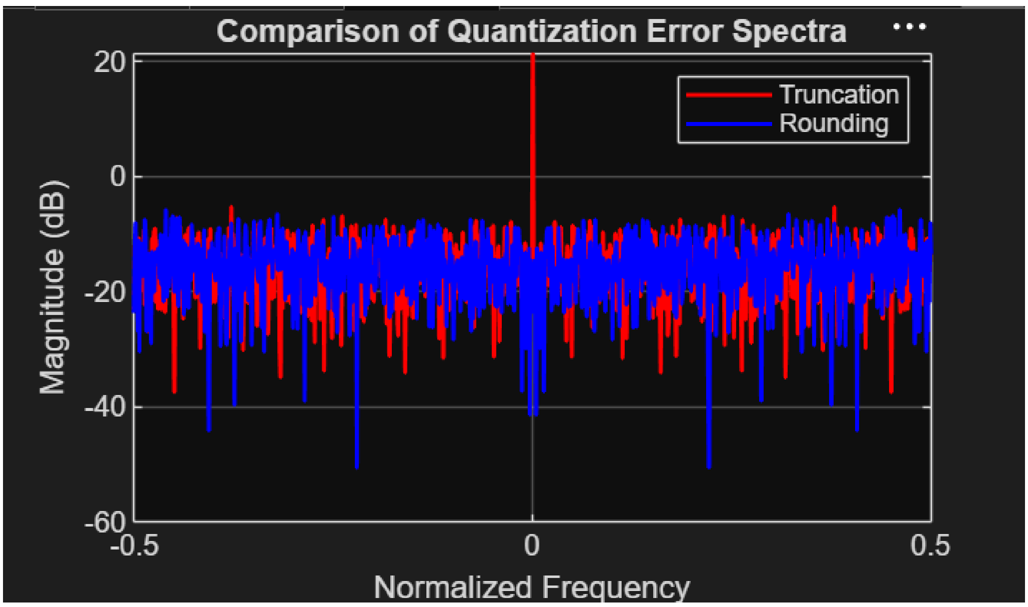
```
legend('Truncation','Rounding')

xlabel('Normalized Frequency')

ylabel('Magnitude (dB)')

title('Comparison of Quantization Error Spectra')

%% ----------------------------

%% Error Statistics

%% ----------------------------

fprintf('\n==============================================\n');

fprintf(' Quantization Error Analysis\n');

fprintf('==============================================\n');

fprintf('Mean Error (Truncation) = %f\n',mean(error_trunc));

fprintf('Mean Error (Rounding)   = %f\n\n',mean(error_round));

fprintf('Variance (Truncation)   = %e\n',var(error_trunc));

fprintf('Variance (Rounding)     = %e\n\n',var(error_round));

fprintf('RMS Error (Truncation)  = %f\n',rms(error_trunc));

fprintf('RMS Error (Rounding)    = %f\n',rms(error_round));

fprintf('==============================================\n');
```

OUTPUT

```
==============================================
Quantization Error Analysis
==============================================
Mean Error (Truncation) = 0.011275
Mean Error (Rounding) = 0.000006

Variance (Truncation) = 4.135769e-05
Variance (Rounding) = 4.314046e-05
RMS Error (Truncation) = 0.012980
RMS Error (Rounding) = 0.006567
==============================================
>>
```


Quantization Error - Truncation
Amplitude
0.02
0.01
0
0
500
1000
1500
2000
2500
3000
3500
4000
4500
Sample Number
Quantization Error - Rounding
Amplitude
0.01
0
-0.01
0
500
1000
1500
2000
2500
3000
3500
4000
4500
Sample Number


Error Spectrum using Truncation
Magnitude (dB)
20
0
-20
-40
-0.5
0
0.5
Normalized Frequency
Error Spectrum using Rounding
Magnitude (dB)
0
-20
-40
-60
-0.5
0
0.5
Normalized Frequency


Comparison of Quantization Error Spectra
Truncation
Rounding
Magnitude (dB)
20
0
-20
-40
-60
-0.5
0
0.5
Normalized Frequency